Street-Litter Detection for Civic Mapping

Does a Detector That Scores Well on TACO Still Give a Map You Can Trust on Real Streets?

Problem	P59: Street-Litter Detection for Civic Mapping [Moderate]
Team	Harshit Periwal (1024031053), Nipun Gupta (1024031169)
Datasets	TACO (five-class subset, as in the baseline paper); a 150-image self-collected local test set, held out
Source model	YOLOv8m (Ultralytics), COCO-pretrained weights, fine-tuned on TACO
Baseline paper	Tran and Hu, TrashDet, arXiv:2512.20746 (2025) — Table 1
Reproduction target	YOLOv8m on the five-class TACO subset: 16.6 mAP50, within 1 point
Map demo	folium (Leaflet) heatmap of geotagged detections along at least two streets
Expected outcome (as set)	mAP on TACO plus a 150-image self-collected local test set; a working map demo showing detection density along at least two streets; precision analysis at deployment thresholds
Deliverable	UCS532 sessional report, CVPR format, 4–6 pages; repository with one-command reproduction script
Duration	7 weeks (Week 3 to Week 9); checkpoint at Week 6

1. Motivation and Central Idea

Street litter is cleaned where someone knows it is. A city that could turn dashcam or phone imagery into a map of where litter accumulates could send crews to the right streets first. The detector is the first stage of that pipeline — image, litter detections, geotag, density on a map — and the map is only as trustworthy as the detections beneath it.

The public benchmark is TACO (Proença and Simões, 2020), an image dataset of litter in the wild — woods, roads and beaches — with COCO-format annotations. Its paper calls it small (1,500 images and 4,784 annotations) and says that satisfactory detection for deployment needs many more annotations; its repository adds that most of the original classes have too few annotations and must be merged or dropped. The most recent paper we found that reports YOLOv8 on TACO (Tran and Hu, 2025) evaluates a five-class subset and gives YOLOv8m 16.6 mAP50. Every one of the eight detectors in its comparison lies between 15.9 and 19.5.

> **The problem with that evaluation.**
>
> *The standard score for a litter detector is mAP50 on held-out images of the same dataset, and that is what the baseline paper reports. Three properties of that number hide what a civic map needs.* ***Same source for train and test:*** *a held-out TACO image comes from the same collection as the training images; a phone photo of one particular local street does not, so mAP on TACO says nothing about how far the score falls where the map will be built.* ***Averaged over thresholds nobody deploys at:*** *mAP50 is the area under the precision–recall curve across all confidence thresholds, but a map runs at one threshold chosen once, and every false positive at that threshold is a false hotspot.* ***Scored per box, consumed per location:*** *the map uses counts at places, not boxes, and box-level mAP does not say whether those counts are faithful.*

The papers we compared (Section 4) report box-level, mAP-style numbers on their own test data. The one that produces a map (Espinosa Juárez et al., 2025) does so on a different dataset, and in the abstract we read the heat maps are presented as an application of the best model. What none of them gives is a number for how much the score changes on unseen local streets, or what precision a fixed threshold delivers there.

> **The central idea of this proposal.**
>
> *We reproduce the baseline paper's YOLOv8m number first, so that nothing is claimed about it before it is trusted. We then treat the map, not the box, as the thing being evaluated and ask three questions with measurable answers: (1) how far does mAP50 on TACO overstate the same detector's mAP50 on our own held-out local street images; (2) if the confidence threshold is chosen using TACO alone, what precision does it deliver on the local streets; and (3) does the detection density on the map agree with the litter actually present, counted by hand? The headline numbers are the gaps between these settings, not the best score.*

2. Objectives

Primary objectives

- **O1** Reproduce the baseline: fine-tune YOLOv8m on the five-class TACO subset and match the published 16.6 mAP50 within 1 point, with the metric code verified against controls before anything else is claimed.
- **O2** Measure the domain gap: evaluate the same trained models, untouched, on the 150-image local test set and report mAP50 and mAP50–95 beside the TACO test result.

- **O3** Precision at deployment thresholds: report precision and recall at fixed confidence thresholds (0.25, 0.50, 0.75) and at a calibrated operating point τ*, on TACO and on the local set, and the transfer loss between them.
- **O4** A working map demo: aggregate geotagged detections into a folium heatmap and show detection density along at least two streets.

Secondary objectives

- **O5** Ablation: how much the result depends on class granularity (five classes versus one "litter" class) and on model size (YOLOv8m versus YOLOv8n).
- **O6** Is the map right? Compare detected counts with hand-counted litter per street segment on the local set.

Objective	Why it is not already answered by the starting point
O1	The baseline paper gives a number but not its split, subset size, class mapping or YOLOv8m training details (Section 4). Reproducing it means fixing those choices openly and reporting how much they matter.
O2	Every paper compared reports results on its own test data. Our local streets are not in any of them, so the gap can only be measured on a set we collect.
O3	mAP50 is threshold-free; a deployed map is not. Precision at one chosen threshold, and whether it survives the move to local streets, is not what mAP50 reports.
O4	The map is the stated end product of P59 and has to be demonstrated, not assumed.
O5	TACO's own repository says rare classes must be merged. How much that choice moves the result is measurable.
O6	Box-level metrics do not test whether per-location counts are faithful; this check matches how the map is used.

3. Novelty

The novelty is not a new detector. It is an evaluation that matches how the detector's output will be used, built on a reproduced published baseline.

- **N1. The gap is the reported statistic.** Pre-declared direction: mAP50 on the local set is lower than on TACO test. The headline number is the difference. If the local score is not lower, that is reported as the finding.
- **N2. Operating point, not area under the curve.** τ* is chosen on TACO validation images only, then applied unchanged to the local set. The change in precision is the transfer loss.
- **N3. A map-level check against hand counts.** Detected counts per street segment are compared with manual counts, so the map is validated in the unit it is displayed in.
- **N4. A genuinely held-out local set.** The 150 local images are never used for training, threshold selection or hyper-parameter choice; TACO validation images do all the tuning.
- **N5. Controls before results.** Oracle, shuffle and annotation-audit checks are built first and must pass before any mAP number is believed.

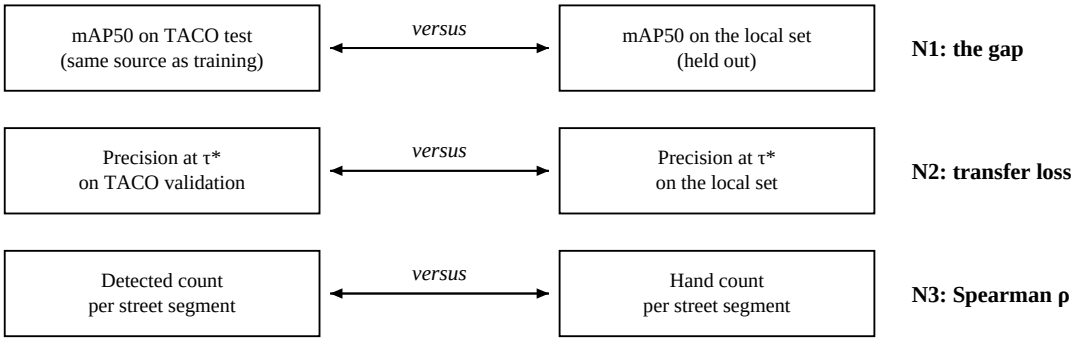


Figure 1. The three comparisons this study reports. Each pairs a TACO-side number with a local-side number and reports the difference.

4. Choosing the Baseline Paper and Reference Numbers

The problem statement fixes the starting point as TACO and YOLOv8. We compared the four papers we found that come closest to it and chose the one whose number we can actually try to reproduce. Everything below was read from the papers' pages or abstracts before this proposal was written.

Tran and Hu, TrashDet (arXiv:2512.20746, December 2025). Uses a five-class TACO subset — paper, plastic, bottle, can, cigarette — as its main benchmark. Its Table 1 compares eight detectors, including YOLOv5m and YOLOv8m, by mAP50, defined as average precision at IoU 0.5 (the area under the precision–recall curve). YOLOv8m is listed at 25.9M parameters and 16.6 mAP50. The paper does not state the number of images or annotations in the subset, the train/validation/test split, how its "paper" and "plastic" classes map onto TACO's 60 categories, or how the YOLOv8m row was trained.

Espinosa Juárez et al. (IEEE CIETES 2025). Evaluates YOLOv8l, YOLOv9c and YOLOv10l on pLitterStreet, a street-litter dataset. YOLOv8l reaches 72% mAP50 on a four-class scheme, and the best model is then run over Google Street View images from eight cities to draw heat maps. It is the nearest precedent for the map half of P59, but it is not TACO, so it cannot be the reproduction target; we use it for positioning only.

Lim et al. (IEEE ICCKE 2023). Trains YOLOv4, YOLOv5 and Faster R-CNN on TACO and reports YOLOv5x as the best, with two figures (88% and 71.4%) whose metric names were not legible on the page we could read. It does not include YOLOv8.

Wadhwanwala and Kumar (Preprints.org, July 2025, not peer-reviewed). Reports YOLOv8 at 84.1% mAP50 on a 970-image, eight-class set (849/81/40 split) built from TACO, a second dataset and manually web-scraped images. The scraped images cannot be recreated, so the number cannot be reproduced from TACO.

Choice. We take Tran and Hu as the baseline paper. It is the only one that matches the starting point exactly (TACO and YOLOv8), states its metric, and gives a number on a subset we can rebuild from the public TACO release. Its YOLOv8m mAP50 of 16.6 is our reproduction target. Espinosa Juárez et al. shape the map and the street-level framing; the other two are positioned in the related work but not reproduced.

Table 1: Published reference numbers to be matched (TACO five-class subset, mAP50, Tran and Hu Table 1)

Method	Parameters	mAP50	Role in this study
YOLOv8m	25.9M	16.6	Our reproduction target (O1)
YOLOv5m	21.2M	15.9	Context; not reimplemented
TrashDet-l (the paper's own method)	30.5M	19.5	Upper reference; not reimplemented

Three observations follow and shape the design. First, all eight detectors in the table sit between 15.9 and 19.5 mAP50, a spread of 3.6 points. A difference of a few points is therefore the scale of the benchmark itself, and any seed-to-seed or split-to-split variation of that order must be shown, not hidden (Section 6.5). Second, because the split, subset size and class mapping are not given, a match means matching the protocol as far as it is stated. We fix the unstated parts openly, vary the split across three seeds, and report the spread. Third, Table 1 prints the same value, 16.6, for YOLOv8m's average recall and its mAP50, which may be a table error. We use only mAP50 and do not treat recall as a target.

Three structural properties that constrain the design

- **TACO is small and long-tailed.** Few annotations per rare class means per-class AP is unstable; this is why the baseline paper uses five classes and why class granularity is an explicit ablation (O5).
- **TACO scenes are not street-level phone or dashcam views.** They include woods and beaches. A domain gap on local streets is expected; it is the quantity O2 measures.
- **150 images is small.** Classes with fewer than 20 local instances get counts only, not per-class AP. Every local result is reported with its image and instance counts. (20 is a design choice fixed here in advance.)

5. Methods

5.1 What the system outputs

For each image the pipeline emits detections (box, class, confidence). Detections with confidence at or above τ are kept; their count and the image's GPS position become one point on the map. τ^* is selected by a function that sees only TACO validation images; the local-set labels are first read by the evaluation script, after the configuration is frozen.

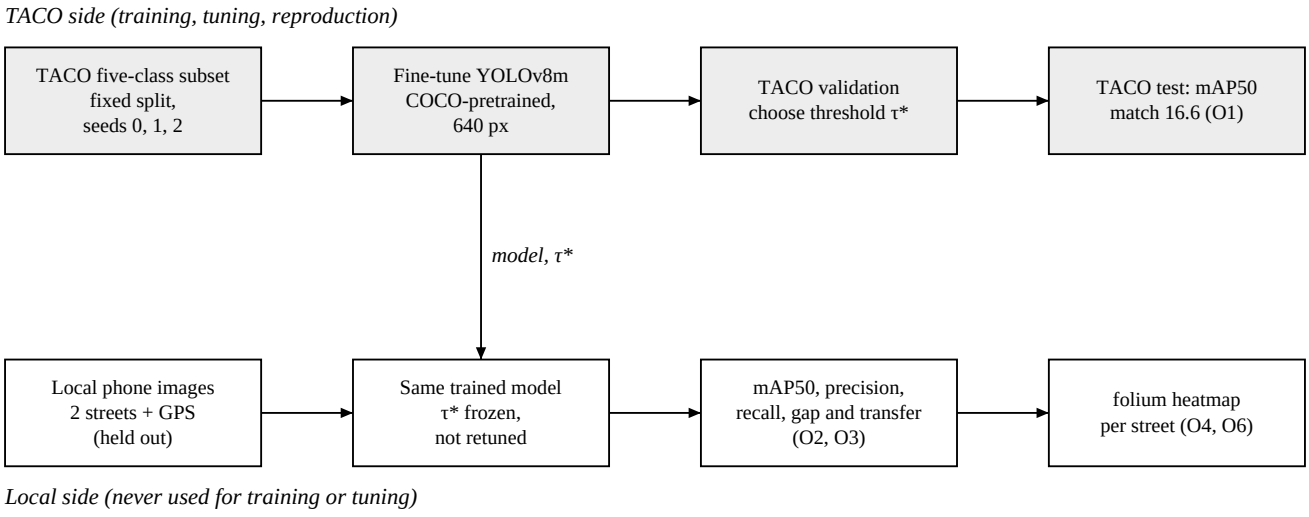


Figure 2. The pipeline. The TACO side trains, tunes and reproduces the baseline; the local side is only ever evaluated.

5.2 Data preparation and protocol choices the paper leaves open

- **Five-class subset.** Built from the public TACO annotations by keeping annotations of paper, plastic, bottle, can and cigarette. The paper does not say how "paper" and "plastic" map onto TACO's categories; we fix one documented mapping in the config after inspecting the annotation file, and list it in the report.
- **Splits.** Three fixed random image-level splits, one per seed (0, 1, 2), because the paper gives none; each seed also fixes the training run. The split files are stored in the repository.
- **Training.** YOLOv8m from COCO-pretrained weights at 640 px (the resolution the paper uses for its own models), Ultralytics defaults otherwise; epochs are fixed in the config before any evaluation.
- **Single-class setup (arm B).** All TACO annotations merged into one "litter" class.
- **Local set.** About 75 images per street on two streets, one phone at a consistent height and orientation in daylight, GPS recorded, including stretches with little or no litter. Every litter item is boxed once and labelled as one of the five classes or as "other litter". Five-class mAP uses the five classes; class-agnostic metrics and the map use all litter. 20 images are re-annotated a week later to measure our own consistency.
- **Consent and privacy.** The camera is aimed at the ground and kerb, not at people. Any image showing an identifiable person is retaken or discarded unless written consent is held. Map output is at street level; no private premises are identified.

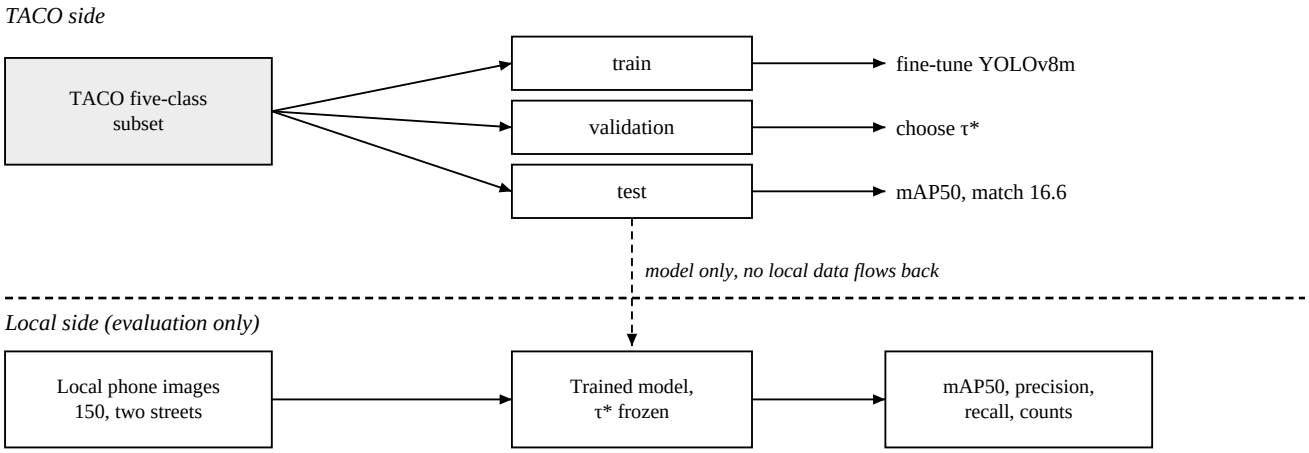


Figure 3. Which data is used for what. Local images are only ever evaluated.

5.3 Metrics and the operating point

Detections are matched to ground truth at IoU ≥ 0.5, greedily by confidence. At a threshold τ:

$$Precision(\tau) = TP / (TP + FP), \quad Recall(\tau) = TP / (TP + FN). \qquad (1)$$

The calibrated operating point τ* is the confidence at which F1 (the harmonic mean of precision and recall) is highest on TACO validation images. It involves no target value chosen by us. Precision and recall are also reported at fixed τ = 0.25,

0.50 and 0.75. For O6, each street is split into consecutive segments of about 10 images (roughly 15 segments in all), and the Spearman rank correlation ρ is computed between detected counts at τ* and hand-counted litter per segment.

5.4 Ablation — this is the actual experiment (O5)

Table 2: Ablation arms

Arm	Configuration	Question it answers
A	YOLOv8m, five-class subset (the baseline)	The reproduction (O1) and the main result for O2–O3.
B	As A, one “litter” class	How much does class granularity change mAP, precision and the map?
C	As A, YOLOv8n	Is the domain gap about capacity, or about data?

Each arm uses three fixed seeds (0, 1, 2) and is evaluated on TACO test and on the local set. Per-seed values are shown, not only means. If seed spread exceeds the gap between arms, that is the result and it is reported as such.

6. Evaluation Protocol

6.1 Metrics

Table 3: Metrics

Metric	Definition	Why
mAP50, mAP50–95	COCO-style, per arm, on TACO test and on the local set	Required by the expected outcome; mAP50 is the baseline paper’s metric.
Gap	mAP50 (TACO test) − mAP50 (local)	N1: the headline statistic.
Precision, recall at τ	Eq. (1) at τ = 0.25, 0.50, 0.75 and τ*	Required “precision analysis at deployment thresholds”.
Transfer loss	Precision(τ*, TACO val) − Precision(τ*, local)	N2: does the chosen threshold hold on local streets?
False positives per image	FP count at τ* on local images with no litter	Directly measures false hotspots.
Count agreement ρ	Spearman, detected vs hand-counted, per segment	O6: is the map faithful?

mAP alone is never the headline; it always appears beside precision at a threshold.

6.2 Pre-declared expected direction

Fixed before any result is seen: (i) the mean TACO-test mAP50 of arm A over the three seeds lies within 16.6 ± 1.0; (ii) mAP50 on the local set is lower than on TACO test; (iii) precision at τ* is lower on the local set than on TACO validation; (iv) detected counts correlate positively with hand counts, with the criterion $\rho \geq 0.5$ (a design choice). If any of these fails, that is reported as the finding and the analysis turns to why, using the failure cases in Fig. 12.

6.3 Splits, and an honest statement about tuning

Epochs, image size, class mapping and τ* are fixed using TACO training and validation images only and frozen in a version-controlled config before the local set is evaluated. No number from the local set may inform any of them. The reproduction is judged on the pre-declared configuration; if it misses 16.6, any variation we try to explain the miss is listed in the report as a variation, and the pre-declared result is still the one reported.

6.4 Mandatory negative controls

Oracle control

Feed the ground-truth boxes in as predictions with confidence 1. mAP and precision must come out at 1.0; anything else is a metric or label-format bug.

Shuffle control

Re-attach predictions to the wrong images. mAP must collapse to near 0. This is the indexing-error detector.

Annotation audit

Draw the converted ground-truth boxes on 20 random images from TACO and 20 from the local set and inspect them. Box coordinates are the usual silent failure when converting COCO to YOLO format.

No-leakage check

Confirm by file hash that no local image appears in any TACO split, and that no image sits in two TACO splits.

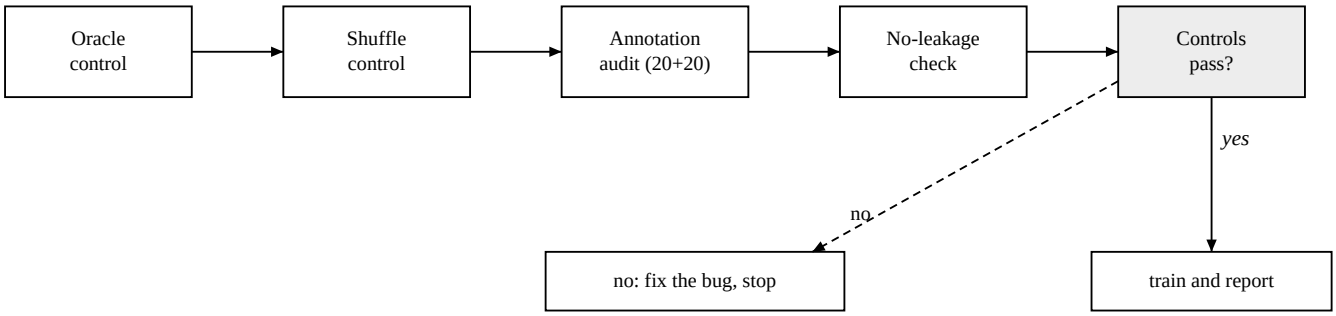


Figure 4. The controls run in order; if any fails, no model is trained until the bug is fixed.

6.5 Stratified reporting

Every headline result is also reported per arm, per street, per seed and per threshold. Failure cases (false positives and misses on the local set) are shown as image examples, not only as counts.

7. Work Plan and Gates

Each phase ends with a gate: a concrete pass/fail test. The next phase does not begin until it passes, so an error in Week 4 is found in Week 4 rather than Week 9.

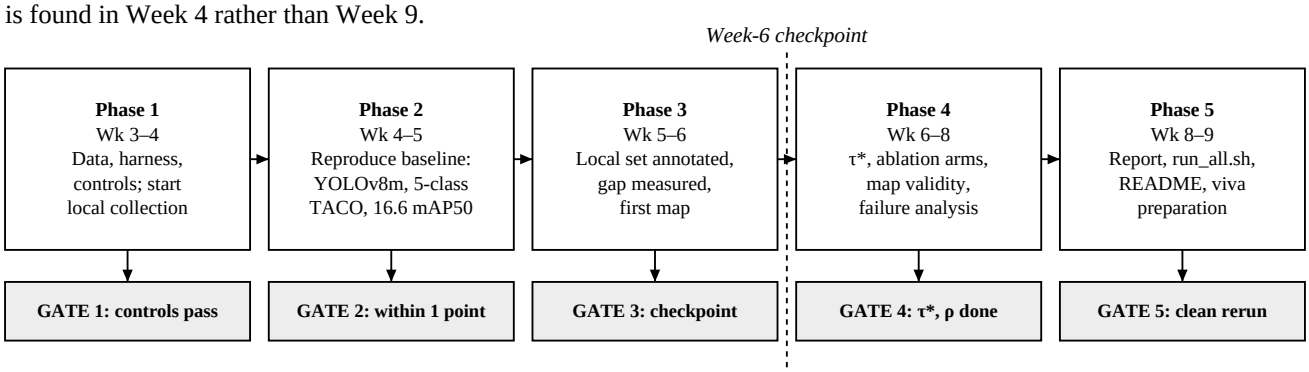


Figure 5. The five phases and their gates across Weeks 3–9, with the Week-6 checkpoint.

Phase 0 (Audit)

GATE 0: PASSED

- Baseline paper chosen after comparing four candidate papers; its Table 1 number and metric definition extracted (Section 4)
- TACO repository and paper checked: size, annotation format, hosting, class-merging note, unofficial-data caveat
- YOLOv8 training interface, COCO-to-YOLO converter and AGPL-3.0 licence confirmed from Ultralytics documentation
- Discrepancy noted: the baseline paper gives no split, subset size, class mapping or YOLOv8m training details

Phase 1 (Data, harness and controls) [Week 3–4]

Build the label conversion, the metric code and the controls before any model is trained. Start collecting local images now, because weather and daylight cannot be rescheduled.

GATE 1

- TACO downloaded and counted; the five-class subset built and its image and annotation counts recorded
- Oracle control returns mAP = 1.0; shuffle control returns near 0; annotation audit passed on 20 + 20 images
- Three splits and the class mapping written to the config; leakage check passed
- folium renders a heatmap from a handful of dummy GPS points
- Local collection protocol fixed; images from at least one street collected with GPS

Phase 2 (Reproduction) [Week 4–5]

GATE 2

- Arm A trained on all three splits; mean mAP50 on TACO test within 1 point of 16.6, per-seed values shown
- If it is not, the discrepancy is localised before proceeding (class mapping, split, epochs), and every variation is recorded
- If the model scores at chance, suspect labels or conversion before claiming a negative result

Phase 3 (Local set and first map) [Week 5–6]

GATE 3 [Week-6 checkpoint]

- All 150 local images collected, GPS-tagged and annotated; consistency re-annotation done
- Arm A evaluated on the local set: mAP50 and the first gap number
- First folium map on two streets from arm A detections
- Repository link submitted with the reproduction table, control outputs and the first map

Phase 4 (Thresholds, ablation and map validity) [Week 6–8]

GATE 4

- τ^* chosen on TACO validation only, frozen, then applied to the local set; precision, recall and transfer loss reported
- Arms B and C run with three seeds each; per-seed spread shown
- Detected-versus-hand-counted comparison per segment, with ρ
- Failure analysis: false positives and misses on local images, grouped by cause

Phase 5 (Report and reproduction) [Week 8–9]

GATE 5

- run_all.sh regenerates every table, figure and the map from a clean checkout
- 4–6 page CVPR-format report; README cites the baseline paper, TACO and YOLOv8 code; AI-assistance declaration complete
- Both team members can explain any part of the code and defend every number (viva)

8. Planned Figures

Figs. 1–5 are the diagrams above. The result figures, each producible from the Phase 2–4 outputs, are:

- **Fig. 6:** Reproduction. TACO-test mAP50 for arm A per seed, against the published 16.6 with the ±1 band marked.
- **Fig. 7:** The gap. mAP50 and mAP50–95 on TACO test versus the local set, per arm, with per-seed points.
- **Fig. 8:** Precision–recall. Curves on TACO and on the local set with $\tau = 0.25, 0.50, 0.75$ and τ^* marked.
- **Fig. 9:** Threshold transfer. Precision versus τ on TACO validation and on the local set; the vertical difference at τ^* is the transfer loss.
- **Fig. 10:** The map. folium heatmap with detection density along at least two streets, with a per-street summary.
- **Fig. 11:** Counts. Detected versus hand-counted litter per segment, with ρ.
- **Fig. 12:** Failure cases. True positives, false positives and misses from the local set, grouped by cause.

9. Timeline

Phase	Weeks	Ends at gate
0 — Audit	complete	Passed
1 — Data, harness, controls	3–4	Controls pass; splits fixed; collection started
2 — Reproduction	4–5	16.6 mAP50 matched within 1 point
3 — Local set and first map	5–6	Gap measured; Week-6 checkpoint submitted
4 — Thresholds, ablation, map validity	6–8	Precision analysis and ρ computed
5 — Report and reproduction script	8–9	run_all.sh runs from a clean checkout

If time runs short, drop in this order: arm C (YOLOv8n); then the consistency re-annotation; then arm B. Never dropped (they are the expected outcome set for P59): the reproduction, mAP on TACO and on the local set, precision at deployment thresholds, and the map on two streets.

Compute

A free-tier Colab GPU or a single GPU is expected to be sufficient: fine-tuning from pretrained weights on a TACO subset, nine runs in total. Every script saves per-run outputs and skips completed runs, so an interrupted session costs little.

10. Risk Register and Contingencies

Risk	Response
Oracle or shuffle control fails	Stop. Find the label-conversion or indexing error before any training.
Reproduction misses 16.6	Report the per-seed spread first (the paper's split is unknown). Then check class mapping, epochs and image size, listing every variation tried. Never replace the pre-declared result with the best variant.
Some TACO image links are dead (images are Flickr-hosted)	Use the Zenodo copy named in the TACO README; record the final counts and use them everywhere.
Near-duplicate TACO scenes land on both sides of a split	Inspect the splits; re-split by scene or batch if leakage is visible, and re-run affected results.
Phone images carry no GPS	Record positions by hand at known points along the street and interpolate; state this in the report.
Local mAP is not lower than TACO mAP	A finding, not a failure: report it and examine why.
Seed spread exceeds the gap between arms	Report the spread and state that the comparison is underpowered at three seeds. Never report the best seed alone.

Limitations stated up front

- The local set is 150 images from two streets taken by one phone in daylight. Nothing here generalises to other cities, cameras, night imagery or true dashcam video, and no such claim is made.
- The reproduction target comes from a paper that omits its split and training details, so a match is a match of the stated protocol only.
- The map shows where the detector fires, not a measured amount of litter; its validity is tested only against our own hand counts.
- Three seeds is few; per-seed values are shown rather than hidden behind means.
- $\rho \geq 0.5$ and the 20-instance minimum are design choices made in advance, not derived values.

11. Expected Contributions

- **C1.** A reproduction of a published YOLOv8m result on TACO with its unstated protocol choices made explicit, and the spread across splits reported.
- **C2.** A measured domain gap: the same detector scored on TACO and on a held-out local street set.
- **C3.** Precision at deployment thresholds, with the threshold chosen on TACO alone and its transfer loss to local streets reported as a number.
- **C4.** A working folium heatmap on at least two streets, validated against hand counts rather than only displayed.
- **C5.** A single-command, gated pipeline from download to figures, with controls built in rather than added at review.

12. Key References

1. T. Tran and B. Hu, "TrashDet: Iterative Neural Architecture Search for Efficient Waste Detection," arXiv:2512.20746, 2025. (The baseline paper; Table 1 is the reproduction target.)
2. P. F. Proença and P. Simões, "TACO: Trash Annotations in Context for Litter Detection," arXiv:2003.06975, 2020. (The dataset.)
3. G. Jocher, A. Chaurasia, and J. Qiu (Ultralytics), "Ultralytics YOLOv8," 2023, github.com/ultralytics/ultralytics. (The detector; AGPL-3.0.)
4. A. Espinosa Juárez et al., "Litter detection in Real-World Environments using YOLO models," IEEE CIETES, 2025. (Street-litter detection and heat maps; positioning.)
5. Z. X. Lim et al., "Leveraging the Power of Object Detection Models in Identifying Litter for a Significant Reduction in Environmental Pollution," IEEE ICCKE, 2023, pp. 265–270. (YOLOv5x on TACO; positioning.)

6. A. Wadhwanwala and T. Kumar, “Revolutionizing Waste Management: Leveraging YOLOv8 for Enhanced Waste Categorization,” Preprints.org, July 2025, not peer-reviewed. (YOLOv8 on a mixed set; positioning.)
7. J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, “You Only Look Once: Unified, Real-Time Object Detection,” in Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR), 2016. arXiv:1506.02640. (The detector family.)
8. T.-Y. Lin et al., “Microsoft COCO: Common Objects in Context,” in Proc. Eur. Conf. Comput. Vis. (ECCV), 2014. arXiv:1405.0312. (Annotation format and mAP protocol.)

Code and tooling

- github.com/pedropro/TACO — dataset, download script, COCO-format annotations (MIT toolkit)
- github.com/ultralytics/ultralytics — YOLOv8 training, validation and COCO-to-YOLO converter
- folium (Leaflet) — map demo and heatmap
- Python: ultralytics, numpy, pandas, scipy (Spearman ρ), matplotlib, Pillow (EXIF GPS)